

Exact norms for the general quadratic Fock evolution operator

A literature-implied answer to Remark 4.6 of arXiv:1409.1262

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Status. `literature_implied_answer` (`full exact characterization`). The open exact-norm question in Aleman–Viola is answered by conjugating the operator to the standard Fock space and applying Neretin’s 2011 norm theorem for Gaussian integral operators. The finite-dimensional identification below does not appear in either source; the norm theorem itself predates the question, so this is not claimed as a new theorem.

The source question

Let

$$H_\Phi = \text{Hol}(\mathbb{C}^n) \cap L^2(\mathbb{C}^n, e^{-2\Phi} dL), \quad P = (Mz) \cdot \partial_z,$$

where Φ is a strictly convex real quadratic form. The evolution is

$$e^{-tP}u(z) = u(e^{-tM}z).$$

Remark 4.6 of [1] says that the exact norm is known when the pluriharmonic part of Φ vanishes, while “the question remains open in the general case.”

Use the source paper’s decomposition

$$\Phi(z) = \frac{1}{2}|Gz|^2 - \Re h(z), \quad h(z) = \frac{1}{2}z^T H z,$$

where G is positive definite and $H = H^T$. Put

$$K = G^{-T} H G^{-1}.$$

Strict convexity is equivalent to $\|K\| < 1$ (for example by Takagi diagonalization).

Exact finite-dimensional answer

Fix a time t for which e^{-tP} is bounded, and define

$$A_t = e^{-tM}, \quad B_t = G A_t G^{-1}, \quad C_t = K - B_t^T K B_t,$$
$$S_t = \begin{pmatrix} C_t & B_t^T \\ B_t & 0 \end{pmatrix}, \quad J = \begin{pmatrix} -I_n & 0 \\ 0 & I_n \end{pmatrix}.$$

The matrix S_t is complex symmetric. Let $s_1(t), \dots, s_n(t) \in [0, 1]$ be its Gaussian canonical invariants in the sense of Neretin [2, Chapter 6, Theorem 1.3]. They are finite-dimensional data: Neretin's Theorem 1.4 says that the roots of

$$\det((I_{2n} - S_t^* S_t) - \lambda(J - S_t^* J S_t)) = 0 \quad (1)$$

come in pairs

$$\lambda = \pm \frac{1 - s_j(t)^2}{1 + s_j(t)^2}, \quad j = 1, \dots, n. \quad (2)$$

Thus the $s_j(t)$ can be recovered from a generalized eigenvalue problem of size $2n$.

Neretin's exact norm formula gives

$$\|e^{-tP}\| = \det(I_{2n} - S_t^* J S_t J)^{-1/4} \prod_{j=1}^n (1 + s_j(t)^2)^{1/2}. \quad (3)$$

The positive value of the fourth root is understood. Formula (3) applies to *every bounded time*, including boundary/noncompact cases [2, Chapter 6, Theorem 2.2(b)]. Whenever $\|S_t\| < 1$, the equivalent formula

$$\|e^{-tP}\| = \det(I_{2n} - S_t^* S_t)^{-1/4} \prod_{j=1}^n (1 - s_j(t)^2)^{1/2} \quad (4)$$

is also available. Equations (1)–(4) are an exact, computable characterization involving only the matrices G, H, M, t .

Why the Gaussian theorem applies

The unitary map used in [1, Lemma 2.3] is

$$Uf(z) = |\det G| f(Gz) e^{-h(z)} : H_{|z|^2/2} \longrightarrow H_\Phi.$$

A direct substitution, with $w = Gz$, gives

$$\begin{aligned} U^* e^{-tP} U f(w) &= e^{h(G^{-1}w) - h(A_t G^{-1}w)} f(G A_t G^{-1}w) \\ &= \exp\left(\frac{1}{2} w^T C_t w\right) f(B_t w). \end{aligned} \quad (5)$$

After the harmless scalar normalization of the standard Fock measure, its reproducing formula is

$$f(\zeta) = \int_{\mathbb{C}^n} e^{\zeta^T \bar{u}} f(u) e^{-|u|^2} d\mu(u).$$

Consequently the operator in (5) has kernel

$$\exp\left\{\frac{1}{2} w^T C_t w + w^T B_t^T \bar{u}\right\} = \exp\left\{\frac{1}{2} \begin{pmatrix} w^T & \bar{u}^T \end{pmatrix} S_t \begin{pmatrix} w \\ \bar{u} \end{pmatrix}\right\}. \quad (6)$$

This is exactly Neretin's Gaussian operator $\mathcal{B}[S_t]$, as defined in [2, Chapter 5, equation (1.4)]. Because U is unitary, $\|e^{-tP}\| = \|\mathcal{B}[S_t]\|$, and (3)–(4) follow from [2, Chapter 6, Theorems 1.4 and 2.2].

The two extremal checks are immediate. If $h = 0$, then $C_t = 0$ and the formula reduces to norm one whenever the pure composition operator is bounded, agreeing with the source's orthogonal case. If $B_t = 0$, (5) is rank one and (4) reduces to $\det(I - C_t^* C_t)^{-1/4}$, the direct Gaussian-vector norm.

Stable large-time sharpening

Assume now that $\text{Spec } M \subset \{\Re \lambda > 0\}$. Then $B_t \rightarrow 0$, $C_t \rightarrow K$, and (5) converges to the rank-one operator

$$T_\infty f(w) = f(0) \exp\left(\frac{1}{2} w^T K w\right).$$

The vacuum evaluation $f \mapsto f(0)$ has norm one in the normalized Fock space, while Neretin's Gaussian-vector identity gives

$$\left\| \exp\left(\frac{1}{2} w^T K w\right) \right\| = \det(I - K^* K)^{-1/4}.$$

The operator-norm return estimate in [1, Proposition 4.4 and Remark 4.6] therefore yields the sharper asymptotic

$$\boxed{\|e^{-tP}\| = \det(I - K^* K)^{-1/4} + O\left(t^{R(M)-1} e^{-t\rho(M)}\right).} \quad (7)$$

In particular,

$$\lim_{t \rightarrow \infty} \|e^{-tP}\| = \det(I - K^* K)^{-1/4}.$$

Since $\|K\| < 1$, this limit equals one exactly when $K = 0$, equivalently when $h = 0$. Thus (7) supplies the missing value behind the source's "if and only if" statement.

Scope and search record

This answers the exact operator-norm question for every bounded time and for every strictly convex quadratic Fock weight in the source setup. It does not claim a simpler radical expression for the roots of the $2n$ -dimensional pencil (1), nor a new closed formula for deciding the source paper's already characterized set of bounded times.

All four cheap run indexes were searched; no prior packet was found. Queries included 1409.1262, the paper title, exact norms, and Gaussian operators. A current web search on 12 August 2026 found later work on boundedness, compactness, essential norms, and special weighted composition operators, but no paper explicitly connecting this remark to Neretin's theorem. The source bibliography contains no Neretin, Olshanski, Howe, or Gaussian-operator reference.

References

- [1] A. Aleman and J. Viola, *On weak and strong solution operators for evolution equations coming from quadratic operators*, J. Spectr. Theory 8 (2018), 33–121; arXiv:1409.1262.
- [2] Y. A. Neretin, *Lectures on Gaussian Integral Operators and Classical Groups*, EMS Series of Lectures in Mathematics, European Mathematical Society, 2011; especially Chapters 5–6.